

D190

Introduction to Healthcare IT Systems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A three-physician family practice in a converted Victorian house uses a digital chart system that stores patient records only for its own clinic and cannot send records to the local hospital. The practice manager has run the office for 20 years. What type of system is this?

- A. A personal health record (PHR).
- B. An electronic health record (EHR).
- C. An electronic medical record (EMR).
- D. A health information exchange.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A patient uses an app she set up herself to track her medications, blood pressure readings, and records she downloads from three providers. What type of record is this?

- A. Any record that holds data from multiple providers.
- B. An electronic health record (EHR).
- C. A personal health record (PHR).
- D. An electronic medical record (EMR).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A hospital's IT team, which recently moved into a new data center, turns on automatic logging of every user who opens a patient record and encrypts laptops used by home-health nurses. The CIO was recruited from a bank. Which HIPAA Security Rule safeguard category do these controls represent?

- A. Privacy Rule requirements.
- B. Physical safeguards.
- C. Administrative safeguards.
- D. Technical safeguards.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which control is an administrative safeguard under the HIPAA Security Rule?

- A.** Annual security awareness training for all staff.
- B.** Encrypting stored patient data.
- C.** Any rule written by the compliance office.
- D.** Badge-controlled doors to the server room.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A regional hospital network that recently merged with a rural clinic wants lab results for the same test, such as serum potassium, to be recognized as the same test across both systems. The clinic's lab is in a renovated barn. Which standard should be used to identify the lab tests?

- A.** ICD-10-CM.
- B.** LOINC.
- C.** CPT.
- D.** DICOM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A developer needs a modern, web-based standard so a patient app can request and receive data from a hospital's EHR through an API. Which standard fits?

- A.** SNOMED CT.
- B.** HL7 FHIR.
- C.** LOINC.
- D.** ICD-10-CM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An audit of 500 charts at a hospital that recently won a design award finds that 18% of records have the allergy field left blank. The values that are entered are correct, and the audit was done by two coders who work remotely. Which data quality dimension is the main problem?

- A. Consistency across systems.
- B. Timeliness.
- C. Completeness.
- D. Accuracy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A quality analyst at a health system with a new mobile app needs to count how many diabetic patients had a foot exam last year. Providers at one clinic, which is known for great customer service, record foot exams only in free-text progress notes. Why will these exams be hard to report?

- A. The clinic's customer service focus interferes with documentation.
- B. The data are inaccurate because they are in notes.
- C. All data in the EHR can be reported equally easily.
- D. Free-text notes are unstructured data that cannot be queried reliably without extra processing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hospital planning a new nurse scheduling system holds interviews and workflow observations with charge nurses to list what the system must do. The project is funded from a capital budget and the lead analyst drives an electric car. Which SDLC phase is this?

- A. Implementation.
- B. Design.
- C. Planning.
- D. Analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which activity belongs in the maintenance and evaluation phase of the SDLC?

- A.** Reviewing user tickets and usage data six months after go-live to plan fixes.
- B.** Interviewing users about needs.
- C.** Estimating project cost and feasibility.
- D.** Training users the week before go-live.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Physicians at a hospital that recently upgraded its Wi-Fi override 94% of drug-interaction alerts, and many say the pop-ups are mostly irrelevant. The hospital's pharmacy director has a doctorate. What is the best way to improve the alerts' safety value?

- A.** Tune the alerts using override data, remove low-value ones, and make only high-severity alerts interruptive.
- B.** Add more alerts so fewer interactions are missed.
- C.** Turn off all drug-interaction alerts.
- D.** Require physicians to take a class and keep the alerts as they are.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A nurse at a clinic with a new check-in kiosk records a single blood pressure reading of 152/94. Later, the EHR displays a graph of the patient's readings over six months showing a steady rise. The clinic opened in 2019. What does the graph represent?

- A.** Data.
- B.** Information.
- C.** Wisdom, because the EHR created it.
- D.** Knowledge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A three-physician family practice ~~in a converted Victorian house~~ uses a digital chart system that stores patient records only for its own clinic and cannot send records to the local hospital. ~~The practice manager has run the office for 20 years.~~ What type of system is this?

Target word: system type **Decides:** One organization's chart, not shared.

A	Confusion set	A PHR is controlled by the patient, not the practice.
B	Adjacent category	EHRs are designed to share records across organizations.
C	Correct	An EMR is a digital chart used within one organization.
D	Inverted operation	An HIE moves records between organizations, the opposite of this system.

2. Answer C Concept 1 • transfer

A patient uses an app she set up herself to track her medications, blood pressure readings, and records she downloads from three providers. What type of record is this?

Target word: reverse: PHR **Decides:** Patient-managed and patient-controlled.

A	Absolute language	Control by the patient, not the number of sources, defines a PHR.
B	Inverted operation	EHRs are maintained by providers.
C	Correct	The patient manages and controls it.
D	Adjacent category	An EMR is one provider's chart.

3. Answer D Concept 2

A hospital's IT team, ~~which recently moved into a new data center,~~ turns on automatic logging of every user who opens a patient record and encrypts laptops used by home-health nurses. ~~The CIO was recruited from a bank.~~ Which HIPAA Security Rule safeguard category do these controls represent?

Target word: safeguard category **Decides:** Audit logs and encryption are system features.

A	Confusion set	The Privacy Rule governs use and disclosure; these are Security Rule controls.
B	Camouflage	Answers the data center detail; physical safeguards cover facility and device access.
C	Adjacent category	Administrative safeguards are policies, risk analysis, and training.
D	Correct	Audit controls and encryption are technical safeguards.

4. Answer A Concept 2 • transfer

Which control is an administrative safeguard under the HIPAA Security Rule?

Target word: reverse: control from category **Decides:** Administrative = policies, risk analysis, workforce training.

A	Correct	Workforce training is an administrative safeguard.
B	Inverted operation	Encryption is a technical safeguard.
C	Absolute language	The content of the control, not who wrote it, decides the category.
D	Adjacent category	Facility access is a physical safeguard.

5. Answer B Concept 3

A regional hospital network that recently merged with a rural clinic wants lab results for the same test, such as serum potassium, to be recognized as the same test across both systems. ~~The clinic's lab is in a renovated barn.~~ Which standard should be used to identify the lab tests?

Target word: choose a standard **Decides:** Identifying lab tests and observations.

A	Adjacent category	ICD-10-CM codes diagnoses.
B	Correct	LOINC provides standard codes for lab tests and clinical observations.
C	Confusion set	CPT codes billable procedures and services, not test identity across systems.
D	Dropped element	DICOM is for medical images, not lab results.

6. Answer B Concept 3 · transfer

A developer needs a modern, web-based standard so a patient app can request and receive data from a hospital's EHR through an API. Which standard fits?

Target word: reverse: need an exchange standard **Decides:** API-based data exchange.

A	Confusion set	A clinical vocabulary, not an exchange protocol.
B	Correct	FHIR is the web-based, API-friendly exchange standard.
C	Inverted operation	Codes the content of lab data but does not move data between systems.
D	Adjacent category	A diagnosis code set.

7. Answer C Concept 4

An audit of 500 charts at a hospital ~~that recently won a design award~~ finds that 18% of records have the allergy field left blank. The values that are entered are correct, ~~and the audit was done by two coders who work remotely.~~ Which data quality dimension is the main problem?

Target word: data quality dimension **Decides:** Blank required fields.

A	Dropped element	No second system is compared in the audit.
B	Confusion set	Timeliness concerns when data are available.
C	Correct	Missing values are a completeness problem.
D	Adjacent category	The entered values are correct; the issue is missing data.

8. Answer D Concept 5

A quality analyst at a health system ~~with a new mobile app~~ needs to count how many diabetic patients had a foot exam last year. Providers at one clinic, ~~which is known for great customer service,~~ record foot exams only in free-text progress notes. Why will these exams be hard to report?

Target word: why reporting is hard **Decides:** Free text is unstructured data.

A	Camouflage	Service reputation does not affect data structure.
B	Confusion set	The issue is format, not correctness.
C	Absolute language	Structured fields are far easier to report than free text.
D	Correct	Reporting depends on structured, coded fields.

9. Answer D Concept 6

A hospital planning a new nurse scheduling system holds interviews and workflow observations with charge nurses to list what the system must do. ~~The project is funded from a capital budget and the lead analyst drives an electric car.~~ Which SDLC phase is this?

Target word: SDLC phase **Decides:** Gathering user requirements.

A	Inverted operation	Implementation builds, tests, and deploys the system.
B	Adjacent category	Design specifies how the system will meet requirements already gathered.
C	Confusion set	Planning sets scope and feasibility before detailed requirements.
D	Correct	Requirements gathering is the analysis phase.

10. Answer A Concept 6 · transfer

Which activity belongs in the maintenance and evaluation phase of the SDLC?

Target word: reverse: activity from phase **Decides:** After go-live: support, measure, improve.

A	Correct	Post-go-live support and evaluation is the maintenance phase.
B	Inverted operation	That is analysis, near the start.
C	Confusion set	That is planning.
D	Adjacent category	Training is part of implementation.

11. Answer A Concept 7

Physicians at a hospital ~~that recently upgraded its Wi-Fi~~ override 94% of drug-interaction alerts, and many say the pop-ups are mostly irrelevant. ~~The hospital's pharmacy director has a doctorate.~~ What is the best way to improve the alerts' safety value?

Target word: CDS improvement **Decides:** High override rate signals alert fatigue.

A	Correct	Fewer, targeted alerts reduce fatigue and restore attention to important ones.
B	Inverted operation	More alerts worsen fatigue and override rates.
C	Absolute language	Removes the useful high-severity alerts too.
D	Dropped element	Leaves the root cause, low-value alerts, in place.

12. Answer B Concept 8

A nurse at a clinic ~~with a new check-in kiosk~~ records a single blood pressure reading of 152/94. Later, the EHR displays a graph of the patient's readings over six months showing a steady rise. ~~The clinic opened in 2019.~~ What does the graph represent?

Target word: DIKW level **Decides:** Data organized in context over time.

A	Adjacent category	The single reading is data; the organized graph is information.
B	Correct	Organizing readings into a trend puts data in context.
C	Camouflage	The tool that made it does not decide the level.
D	Confusion set	Knowledge is the interpretation used to act, such as deciding to adjust treatment.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	EMR vs. EHR vs. PHR	An EMR is one organization's digital chart; an EHR is designed to share a patient's record across organizations and care settings; a PHR is managed and controlled by the patient.	Using EMR and EHR as exact synonyms, or calling a patient portal view a PHR the patient controls.	Who maintains the record and whether it is shared beyond one practice.
2	HIPAA Security Rule safeguards	The Security Rule protects electronic PHI through administrative safeguards (policies, risk analysis, training), physical safeguards (facility and device access), and technical safeguards (access control, audit logs, encryption).	Classifying training as technical or encryption as administrative.	A control that is a policy, a physical barrier, or a system feature.
3	Health data standards and code sets	ICD-10-CM codes diagnoses; CPT codes procedures and services; LOINC identifies lab tests and observations; SNOMED CT codes clinical terms; HL7 FHIR exchanges data between systems; DICOM handles medical images.	Using a billing code set to identify lab tests or an exchange standard as a vocabulary.	What kind of thing needs a standard code or format.
4	Data quality dimensions	Key data quality dimensions are accuracy (correct), completeness (nothing missing), timeliness (available when needed), and consistency (same across systems).	Calling any data problem an accuracy problem.	A description of what is wrong with the data.
5	Structured vs. unstructured data	Structured data sits in defined, coded fields that can be queried and reported; unstructured data is free text, scanned documents, or images that need extra processing to analyze.	Assuming anything stored in the EHR is easy to report on.	A reporting need that depends on where the information was entered.
6	Systems development life cycle	Planning (scope, feasibility), analysis (gather requirements from users), design (specify the solution), implementation (build, test, train, go live), and maintenance/evaluation (support, measure, improve).	Jumping to vendor selection or design before requirements are gathered.	An activity in a health IT project that must be matched to a phase.

#	Concept	Rule	Trap	Tell
7	Clinical decision support and alert fatigue	Too many low-value alerts cause clinicians to override alerts, including important ones; effective CDS is targeted, tiered by severity, and tuned using override data.	Adding more alerts to improve safety.	High override rates or complaints about pop-ups.
8	Data, information, knowledge	Data are raw facts; information is data organized in context; knowledge is information interpreted to guide action.	Calling a single raw value “information.”	Whether values are raw, organized, or applied to a decision.

Course scope (public description): Introduction to Healthcare IT Systems introduces students to healthcare information technology as a discipline. Focusing on evaluating health information systems and collecting data, students will learn the various roles and functions of the health information manager in supporting the business of healthcare. This course introduces students to information technology as a discipline. This course also exposes students to the various roles and functions of the health information manager in supporting the business of healthcare. Students will learn through e-text readings, videos, case studies, several modules from LinkedIn Learning, knowledge checks, and unit quizzes. There are no prerequisites for this course.